Enterprise engineering studio · Intelligent software for modern commerce

Intelligent systems built to quietly run commerce.

Sanad constructs operational backbones and bespoke autonomous logic for commercial enterprises. We remove the cognitive and administrative weight of scaling — engineering a structural reduction of at least 30% in daily operational load.

Request Technical Dossier →

Read the 30% baseline

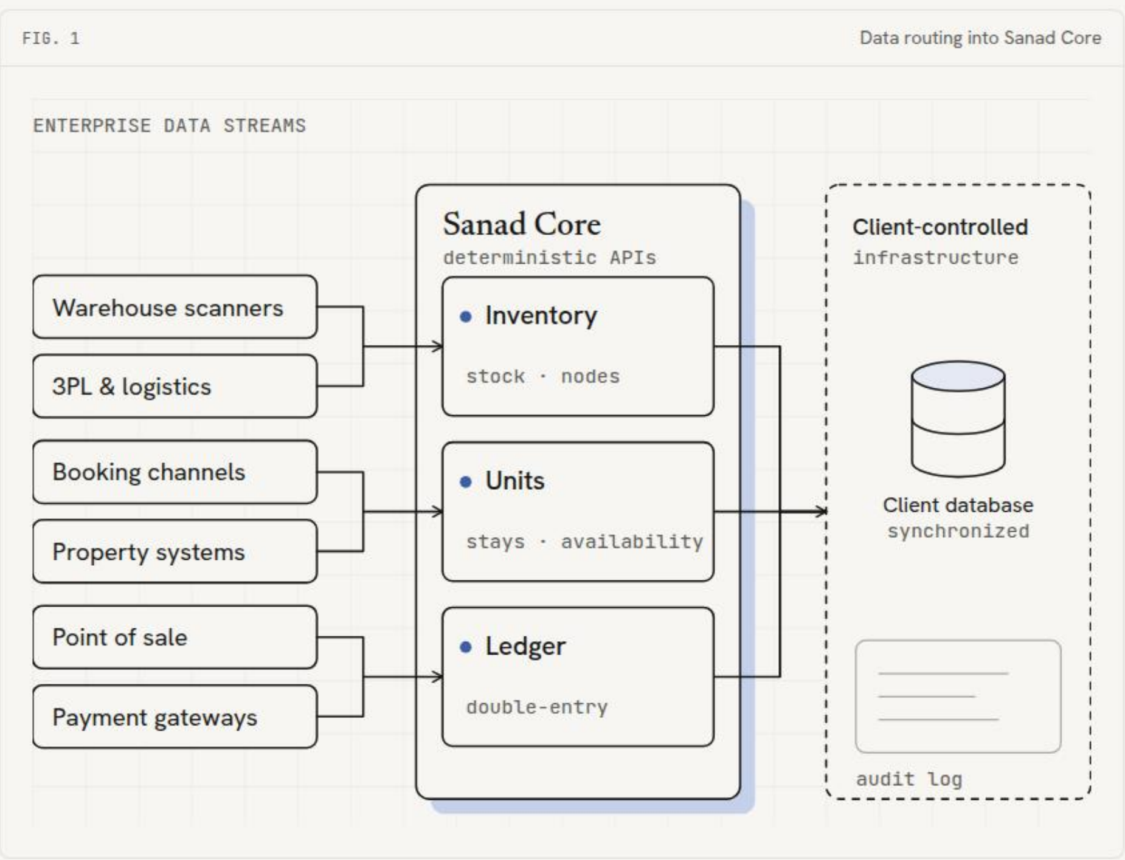
≥ 30%
Structural reduction in daily operational load

18 h → < 1 s
Inventory and supply synchronization

< 100 ms
Internal state change across terminals

Zero
Black-box dependencies

FIG. 1 · Data routing into Sanad Core



Enterprise data streams route into the Sanad Core modules, then synchronize out to infrastructure the client controls.

For Leadership

Predictable velocity, margin protection, and operational clarity.

For Engineering

Decoupled architecture, local-first integrity, and zero black-box dependencies.

01 Proof by metric

The 30% efficiency baseline

Operational load is not a feeling. It is waiting, re-keying and reconciling — three things that can be measured, and removed. Every system we build is specified against one baseline: a structural reduction of at least 30% in daily operational load.

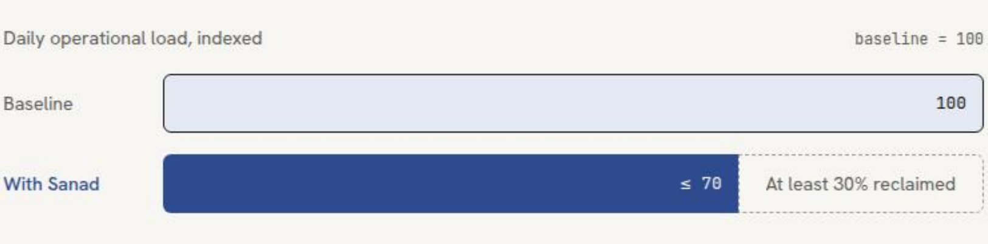
Workstream	Before	With Sanad
01 **Inventory & Supply Sync**	18 hours of multi-channel lag between stock and sales	Sub-second synchronization across every channel
02 **Financial & Reconciled Auditing**	Repeated reconciliations, performed by staff each cycle	Zero-loss balancing routines, automated end to end
03 **Human Error Overhead**	Re-keyed data entered again in each system it touches	Eliminated structurally: one entry, one record of truth

Definition

`L = t_wait + t_rekey + t_reconcile`
daily operational load, in hours

`L′ ≤ 0.70 × L`
the 30% Law

Daily operational load, indexed



02 Systems & architecture

Ready-made modules.
Bespoke engineering.

Two explicit offerings, never blended. Pillar A is production-tested software, configured for your business. Pillar B is engineered from scratch, for operational flows that belong to you alone.

Pre-configured SaaS · Pillar A

A Turnkey Intelligent Modules

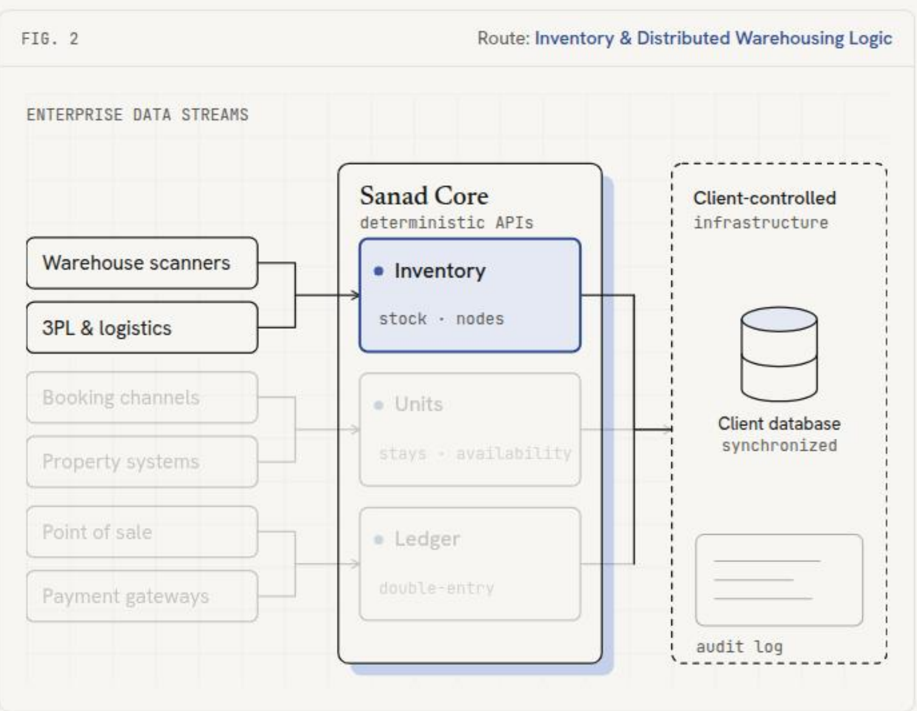
High-efficiency, production-tested systems, ready for deployment.

Select a module. Follow the data.

Choose Inventory, Units or Ledger to trace which sources feed it, which record it owns, and what it sends to your infrastructure.

- A.1 **Inventory & Distributed Warehousing Logic**
  One stock position across every warehouse, store and channel.
- A.2 **Hospitality & Unit Management Engines**
  Units, stays and turnover, held to a single source of availability.
- A.3 **Unified Commerce & POS Ledger Sync**
  Every sale, online or in store, posts once to one ledger.

FIG. 2 · Route: Inventory & Distributed Warehousing Logic



Owns

Stock position per node, reservations and transfers.

Fed by

Warehouse scanners · 3PL & logistics

Schema

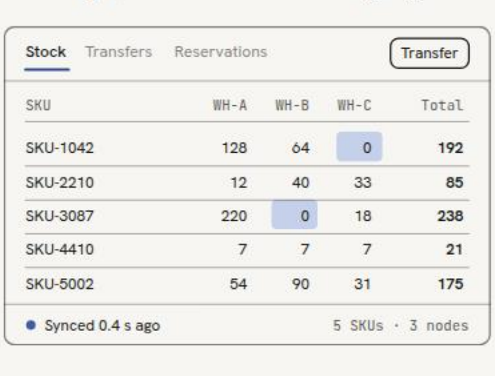
```
type StockLevel {
  sku      string
  node     NodeId  // warehouse or store
  on_hand  int
  reserved int  // held for open orders
  version  lamport  // deterministic ordering
}
```

Parameters

Topology	`n-node, multi-warehouse`
Sync	`event-sourced, deterministic merge`
Reservations	`per-channel holds, auto-expiry`
Offline	`node-local ledger, reconcile on reconnect`
Latency	`< 100 ms in-site, sub-second across channels`

A.1 · Interface wireframe

Inventory & Distributed Warehousing Logic

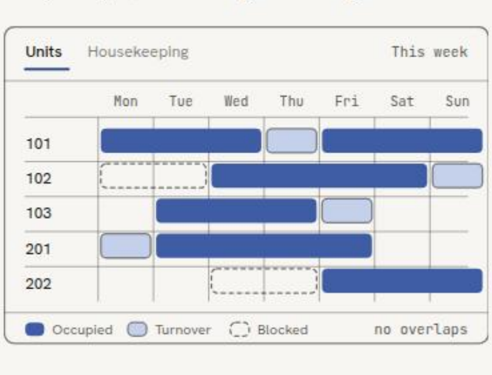


SKU	WH-A	WH-B	WH-C	Total
SKU-1042	128	64	0	192
SKU-2210	12	40	33	85
SKU-3087	220	0	18	238
SKU-4410	7	7	7	21
SKU-5002	54	90	31	175

● Synced 0.4 s ago · 5 SKUs · 3 nodes

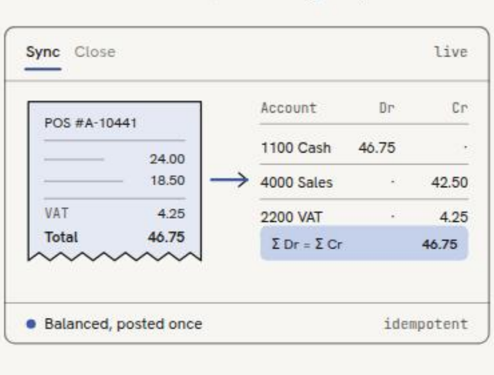
A.2 · Interface wireframe

Hospitality & Unit Management Engines



A.3 · Interface wireframe

Unified Commerce & POS Ledger Sync



POS #A-10441
24.00
18.50
VAT 4.25
Total 46.75

Account	Dr	Cr
1100 Cash	46.75	·
4000 Sales	·	42.50
2200 VAT	·	4.25
Σ Dr = Σ Cr		46.75

● Balanced, posted once · idempotent

Bespoke systems · Pillar B

B Tailored Commercial Architecture

From-scratch systems, engineered for unique, proprietary operational flows.

B.1

Custom internal operating systems

Replacing patchwork spreadsheets

One system of record in place of a folder of spreadsheets: role-based views, enforced workflows, and a history of every change.

record
`single source of truth`

access
`role-based`

history
`append-only`

B.2

Private agentic workflows

Operating securely on client infrastructure

Autonomous logic that runs inside your perimeter, on your hardware or your cloud account. Every action is logged, bounded by explicit permissions, and reversible.

runtime
`client infrastructure`

egress
`denied by default`

oversight
`approval gates, full action log`

B.3

High-availability distributed databases

With offline-first synchronization

Replicated stores that keep accepting writes when the network does not, then converge deterministically when it returns.

writes
`local-first`

merge
`deterministic`

single point of failure
`none, by design`

03 System mechanics & technical rigor

Specified like infrastructure, stated so it can be tested.

For CTOs and technical evaluators

Data ownership
Sovereign
Complete sovereignty. Databases synchronize with client-controlled infrastructure.

Reliability
Offline-capable
Operational nodes designed to function without internet disruption.

Latency
< 100 ms
Sub-100ms internal state changes across multi-terminal setups.

Integrations
Deterministic
Clean, deterministic APIs connecting legacy ERPs, payment gateways, and third-party logistics.

Representative event trace — Illustrative sequence, not a benchmark.

```
# multi-terminal sale on the shop floor
t+000 ms     term-02    sale.created       tx=A-10441
t+008 ms     node-01    ledger.posted      Σdr = Σcr ok
t+019 ms     node-01    stock.reserved     sku=1042 -1
t+037 ms     term-01    state.applied      v=2291
t+041 ms     term-03    state.applied      v=2291

# network lost: the node keeps working locally
t+912 ms     term-02    sale.created       tx=A-10442
t+921 ms     node-01    ledger.posted      queued=1

# network restored: replicate to client-db
t+4.2 s      node-01    replicate          converged v=2292
```

Target: state changes under 100 ms across terminals, with no dependency on the network.

04 Security & compliance

Stated plainly, so it can be checked.

Properties of the systems we build, written as specifications rather than assurances.

We do not claim certifications on this page. If you must meet a specific regulation, we design to it and show you how.

✓ **Data location**	Client-controlled infrastructure.
✓ **Access**	Role-based. Permissions are explicit and reviewable.
✓ **History**	An append-only record of changes and actions.
✓ **Agents**	Bounded permissions, approval gates, a full action log, and every action reversible.
✓ **Network**	Outbound access denied by default.
✓ **Failure**	Offline-capable nodes and deterministic recovery.

05 How we work

Four stages. One owner at each.

Every engagement follows the same order, so you always know what is being decided and what comes next.

01 Understand
We map how work actually moves through your business: systems, hand-offs, waiting and re-entry.

You receive
A written map of current operations, and the baseline for the 30% target.

02 Design
We specify the system: data model, integrations, failure behavior, and what must run offline.

You receive
An architecture document you can hand to your own engineers.

03 Build
We build and test in your environment, in stages, each one usable on its own.

You receive
Working software, tested against the baseline.

04 Operate
We run it with you, monitor it, and hand over ownership when you are ready.

You receive
Runbooks, access, and the option to run it yourselves.

06 Engagement models

Two ways to start.

	Pillar A Turnkey modules	Pillar B Bespoke systems
Best when	Your process matches a standard shape: stock, units, a sales ledger.	Your operations are particular to you and no product fits.
What we do	Configure, connect and deploy a tested module.	Specify and engineer from scratch.
From you	Access to current systems, a named process owner, and your configuration decisions.	Time with the people who run the work, access to systems and data, and a named decision-maker.
You get	A running system, and training.	A system built to your flow, with full documentation.
Start with	The technical dossier.	An architectural walkthrough.

07 Questions from engineers

Plain answers to the usual questions.

Longer answers, with diagrams, are in the technical dossier.

Ask us directly →

Who owns the data? —

You do. Databases synchronize with infrastructure you control, and we keep no copy that we depend on.

What happens without internet? +

How do we migrate from our current system? +

Can it connect to our ERP and payment gateway? +

Do agents send our data to outside services? +

What if we outgrow a turnkey module? +

08

The Sanad manifesto

Software should clarify the mind, not demand it.

Most enterprise software creates its own bureaucracy. Teams spend hours managing the tool that was supposed to save them time. Sanad is built on the inverse thesis: intelligent software works silently in the background so leaders and builders can dedicate their focus to what actually matters.

Support for the people who lead.

If the work is heavier than it should be, we should talk.

Request Technical Dossier →

or write to inquiries@sanad.systems

Sanad

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● Accepting new engagements · Remote-first studio

Documentation
- Technical documentation
- Whitepapers
- Operational blueprints

Shared on request, with the technical dossier.

Inquiries

inquiries@sanad.systems

For an architectural walkthrough of any system on this page, write to us directly.

Support for the people who lead.